

4/8/2025 Welcome and Research Instructions for Superconducting Quantum Processor Experiments (Spring Term)

Overview

Welcome to the Wong's Lab! This document provides an overview of the research activities you will participate in. You will be working with superconducting quantum devices, including bosonic and Transmon qubits, and a traveling wave parametric amplifier (TWPA). Please read through the entire document carefully, as it outlines both the planned experimental timeline and useful background resources to help you get started.

1. Experimental Schedule and Workflow

Current Phase: Bosonic Qubit Chip Experiments

From now until the end of April, our dilution refrigerator will host a bosonic qubit chip. Cody is currently leading experiments on this chip, and the first student group will be working alongside him.

Next Phase: Candle Qubits and TWPA Integration

After the bosonic chip run concludes, we will load a new chip with "candle" Transmon qubits along with a TWPA. During this phase, we will focus on calibrating the system: measuring the performance of the TWPA, understanding the signal chain's actual attenuation and gain, characterizing pulse distortion, and estimating the system's overall noise level.

2. Suggested Experiments

In both phases, students will perform foundational qubit characterization experiments. These include:

- **Punchout measurements**
- **Rabi oscillations**
- **T_1 (energy relaxation time)**
- **T_2 (Ramsey spectroscopy)**

These are similar in spirit to what you may have done in the NMR labs but adapted to the superconducting qubit platform. Once these baseline experiments are complete, and if

you have additional time or curiosity, we encourage you to pursue advanced topics such as:

- **Spin echo experiments**
- **Randomized benchmarking (RB)**
- **Quantum process tomography**
- **Many-body dynamics simulations**
- **Two-tone spectroscopy for qubit-resonator systems**
- **Qubit-resonator dispersive shift extraction**
- **Parametric Amplifier Calibration**

We welcome your curiosity—if you are interested in exploring new directions, feel free to discuss your ideas with us!

3. Recommended Reading

Superconducting qubit systems are often less intuitive than atomic or ion-based platforms, particularly due to the microwave engineering aspects involved. To help you understand the underlying physics and practical considerations, we strongly recommend reading *An Engineer's Guide to Superconducting Qubits* by William D. Oliver and colleagues at MIT. A copy of this document will be attached to the same email.

While the full guide is comprehensive, we suggest beginning with the following key sections:

- **II.A, II.B** – Energy levels and nonlinearity in superconducting qubits
- **IV.A–D** – Qubit control and microwave engineering considerations
- **V.A, V.B, V.E** – Qubit readout and Parametric Amplifiers

Before moving on to working with specific samples or running experiments, we **highly recommend** that you read through these sections. They will provide valuable context for understanding the components you'll be working with, and help you get the most out of your time in the lab.

4. System Overview

Transmon Qubits (MIT SQUILL "Candle" Chip)

The Transmon qubit is one of the most widely used superconducting qubit architectures, known for its simplicity, scalability, and relatively long coherence times. It is formed by a Josephson junction shunted by a large capacitor, creating a weakly anharmonic oscillator that can be used as a controllable quantum bit.

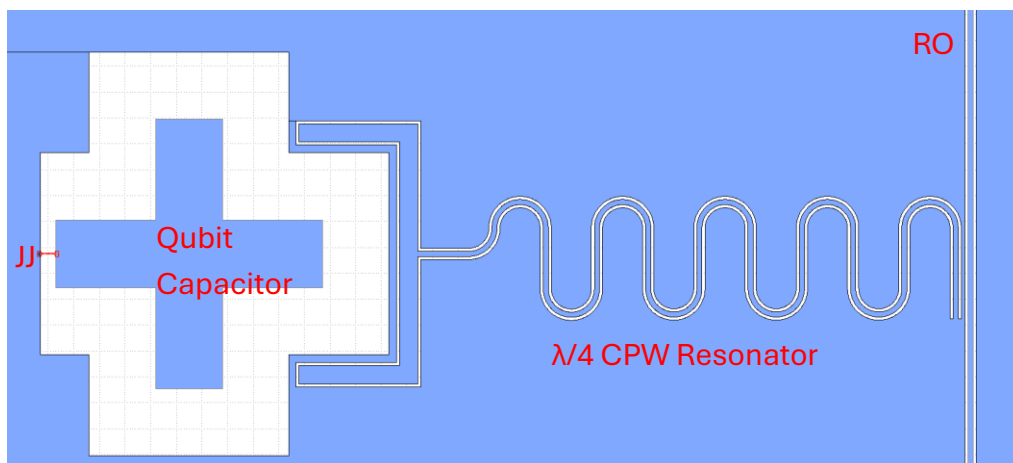
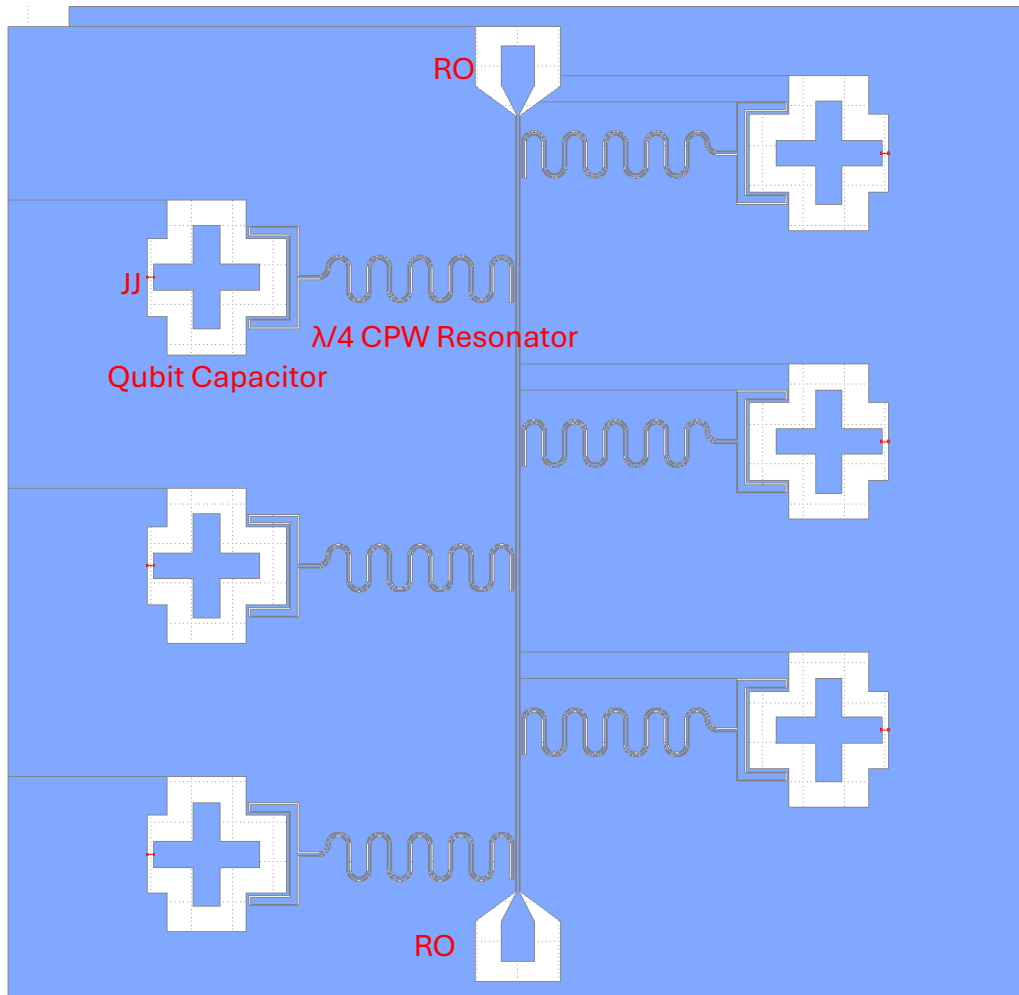
In the chip layout image provided below, you'll find several key components. Here's a guide to interpreting the design:

- **Cross-shaped pads** → These serve as the **capacitor** of the Transmon qubit. They store electric charge and define the qubit's charging energy, which contributes to its discrete energy levels.
- **Josephson junctions (JJs)** → These extremely small elements are not directly visible in the image, but their **locations are marked in red**. They introduce the essential nonlinearity that makes the Transmon a qubit rather than a harmonic oscillator.
- **Readout resonators** → These are **meandering or $\lambda/4$ coplanar waveguide structures** coupled to each qubit. They enable **dispersive readout**, allowing the qubit state to be inferred via shifts in the resonator frequency.
- **Flux bias lines** (*Not present in this design*) → In other Transmon variants, these are used to tune qubit frequency by applying magnetic flux through a superconducting loop containing the Josephson junctions.
- **XY control lines** (*In this design, the XY drive signals are delivered through the readout lines*) → These lines deliver microwave pulses to manipulate the qubit state.
- **Readout lines** → These connect the readout resonators to the output port, where signals are routed to amplifiers (including the TWPA) and digitizers for measurement.

Color interpretation in the layout:

- **Blue regions** indicate **aluminum deposited** on the silicon wafer—this is the superconducting material forming the circuit elements.
- **White regions** indicate where **aluminum has been removed**, exposing the bare **silicon substrate**. These areas form the gaps between structures or isolation regions.

Understanding this layout and color scheme will help you quickly recognize qubit components and signal paths during design review or debugging. Let us know if you'd like to walk through a sample layout together before diving into experiments.



Bosonic Qubits

Bosonic qubits use the quantum states of harmonic oscillator modes—such as microwave cavities or coplanar waveguide resonators—to encode quantum information. These modes can host multiple excitations and exhibit long coherence times, making them ideal for implementing error-correctable logical qubits and exploring fault-tolerant quantum protocols.

In our setup, the bosonic storage mode is realized either in a 3D aluminum cavity or a 2D CPW resonator. A superconducting Transmon qubit is used as an interface, enabling both coherent control of the bosonic state and high-fidelity readout through standard circuit QED techniques. The Transmon mediates interactions between the bosonic mode and the external control/readout circuitry, allowing you to prepare, manipulate, and measure encoded quantum information.

What to look for in the layout or photo:

- Both the **storage mode** and **readout mode** are implemented as **half-wavelength ($\lambda/2$) CPW resonators**.
- A **Transmon qubit** is placed in the **middle** between the storage and readout resonators. Its role is to mediate interactions while mitigating direct cross-talk between the two bosonic modes.
- The **Transmon** has its own dedicated **drive line**, allowing for direct XY control and manipulation of its state.
- The **storage resonator** also has an independent **drive line**, which is used to selectively populate or probe its quantum state.

For a comprehensive overview of bosonic qubits in circuit QED systems, we recommend reading:

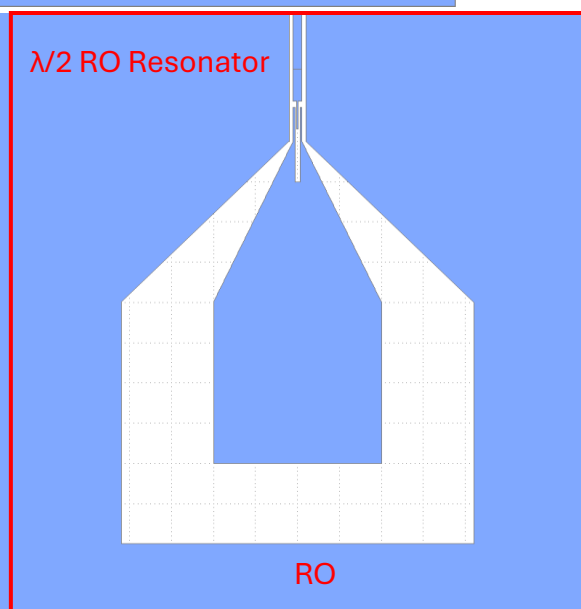
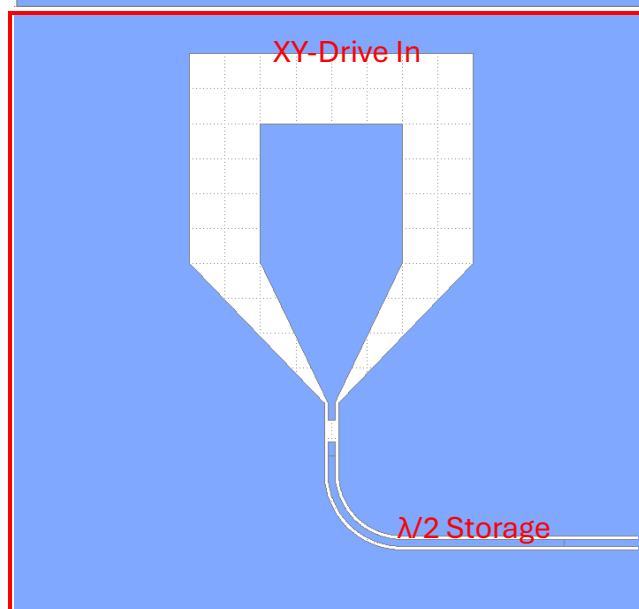
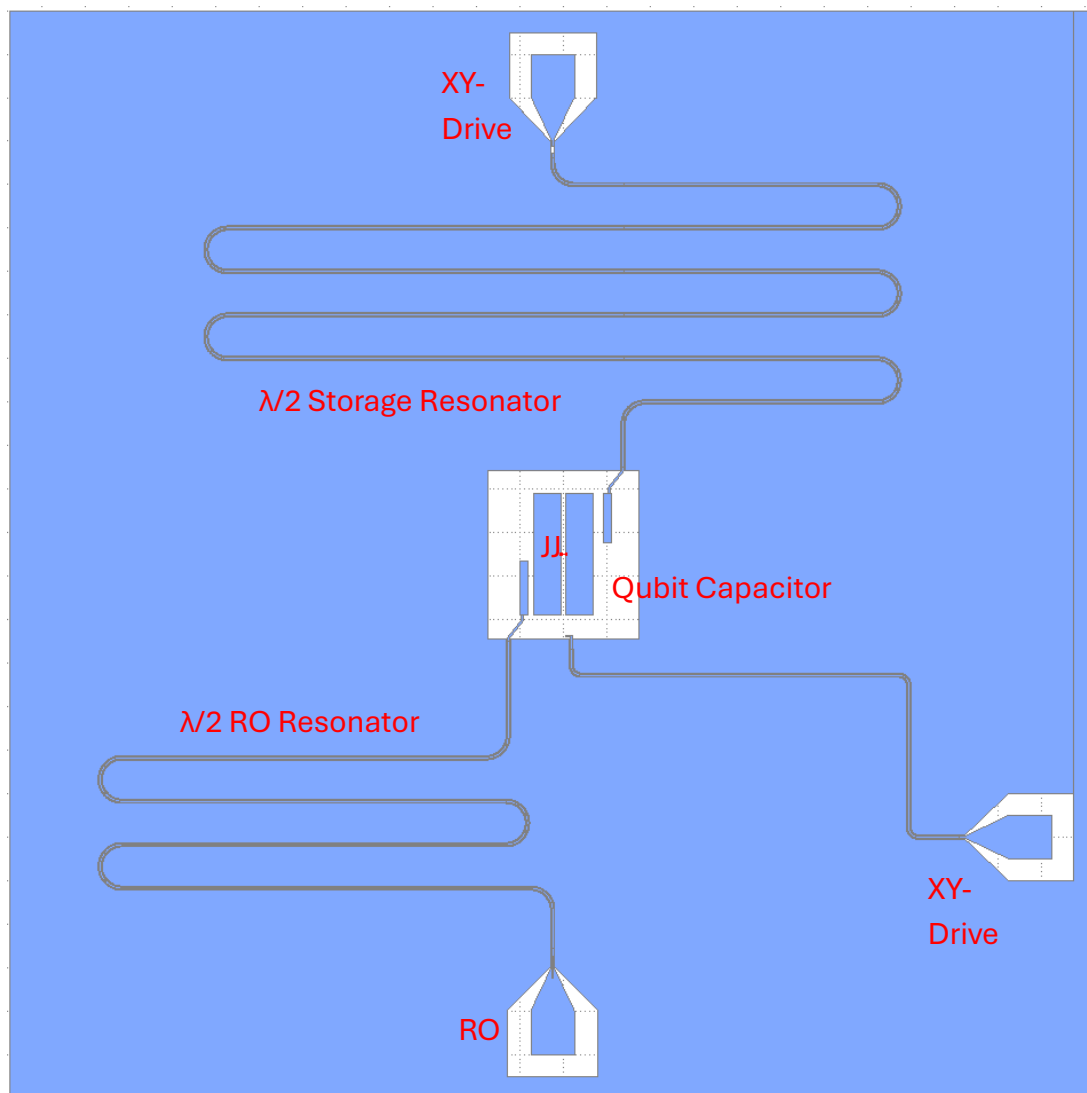
"Quantum information processing with bosonic qubits in circuit QED"

Atharv Joshi, Kyungjoo Noh, and Yvonne Y. Gao

Quantum Science and Technology, **6**(3), 033001 (2021)

DOI: 10.1088/2058-9565/abe989

This review provides both theoretical foundations and experimental context for understanding how bosonic qubits are engineered and controlled in superconducting systems.



Traveling Wave Parametric Amplifier (TWPA)

The TWPA is a broadband, nearly quantum-limited microwave amplifier that plays a crucial role in high-fidelity qubit readout. It consists of a long chain of nonlinear superconducting elements, typically Josephson junctions or SQUIDs, embedded in a transmission line structure. When pumped with a strong microwave tone, the TWPA can amplify weak quantum signals—like the readout tone from a qubit—while adding minimal noise.

In our system, the TWPA is placed at the base temperature stage of the dilution refrigerator and is connected to the readout chain just before the HEMT amplifier. It improves the overall signal-to-noise ratio of the system, enabling faster and more accurate measurements.

What to note about the TWPA hardware:

- It looks like a **long, meandering transmission line** on a chip, often with ground plane cuts and periodic structures.
- **Pump tone** is injected separately and must be carefully calibrated to ensure optimal gain without introducing distortion or instability.
- Calibration involves measuring **gain profiles**, checking **saturation power**, and observing **phase-matching conditions**.

You'll gain hands-on experience by calibrating the TWPA and observing how it boosts measurement fidelity in both single-shot and ensemble averaged experiments.

To understand the operating principles and design of TWPAs in more detail, we recommend reading: **“A near-quantum-limited Josephson traveling-wave parametric amplifier”** by C. Macklin *et al.*, *Science*, **350**(6258), 307–310 (2015).

DOI: 10.1126/science.aaa8525

